Chromatic Admissibility Contexts and Enforced Completeness of Nonadjacent Vertices

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In a d-uniform quantum hypergraph $H = (V, E)$, every measurement context is rainbow in each strong d-coloring. We study *chromatic admissibility contexts*: sets of d pairwise nonadjacent vertices that receive all d colors in every strong d-coloring, despite representing pairwise nonorthogonal rays in a faithful orthogonal representation. Equivalently, their two-valued-state sum is exactly one on the extendable state set $\mathcal{S}_{\mathrm{ext}}(H)$. The broader notion of an admissibility context relative to a specified state family also permits a *two-valued admissibility context*, requiring exclusivity and completeness on all of $\mathcal{S}(H)$. This single-set normalization differs from Navara–Svozil pseudocontexts, which equate two probability sums through a balanced affine certificate. We prove that such certificates imply color-multiset equivalence and distinguish this paired relation from admissibility, universal rainbow exclusivity, and chromatic twins. Within the examined Yu–Oh restrictions, pairwise-nonorthogonal chromatic admissibility triples first occur at $H_{20,12}$ (2 triples) and $H_{21,13}$ (21 triples, classified by even-permutation transversals of its color-signature grid, including a coplanar triple and a 60° triple). We then construct faithfully coordinatized real three-dimensional hypergraphs ($G_{41,25}$ and $G_{41,26}$) possessing a genuine *two-valued admissibility context* $U_{\mathrm{TVS}} = \{3, 27, 28\}$ of three pairwise nonorthogonal rays whose state sum equals one across *all* admissible two-valued states $\mathcal{S}(H)$, and show how attaching 1–0 gadgets promotes the chromatic admissibility contexts of $H_{20,12}$ and $H_{21,13}$ to two-valued admissibility contexts. Operator rigidity remains valid because $\{0, 1\}$-integrality on Specker bugs causes $\mathrm{aff}\,\mathcal{S}(H) \subsetneq \mathcal{A}(H)$, separating two-valued completeness from affine or operator completeness.

I. INTRODUCTION

A complete rank-one projective measurement in $\mathbb{F}^d$ ($\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$) is represented by an orthonormal basis of rays $\{[q_v] : v \in e\}$ whose rank-one projectors $P_v = q_v q_v^\dagger / \|q_v\|^2$ resolve the identity,

$$\sum_{v \in e} P_v = I_d. \tag{1}$$

For any finite d-uniform hypergraph $H = (V, E)$ of intertwined projective measurements ($n = |E|$), every admissible probability assignment $p : V \to [0, 1]$—whether generated by the quantum Born rule $p(v) = \mathrm{Tr}(\rho P_v)$, by convex combinations of classical two-valued states $s : V \to \{0, 1\}$ in $\mathcal{S}(H)$, or by more general normalized weight assignments—satisfies the context normalization equations

$$\sum_{v \in e} p(v) = 1 \qquad (\forall\, e \in E). \tag{2}$$

We write $\mathcal{W}(H)$ for all normalized weights $p : V \to [0, 1]$ satisfying Eq. (2), and $\mathcal{S}(H) = \mathcal{W}(H) \cap \{0, 1\}^V$ for its two-valued states. The entire real affine solution space is

$$\mathcal{A}(H) = \Big\{x \in \mathbb{R}^V : \sum_{v \in e} x(v) = 1 \text{ for all } e \in E\Big\}. \tag{3}$$

These three domains must be distinguished: an identity on $\mathcal{S}(H)$ need not be an identity on $\mathcal{W}(H)$ or on $\mathcal{A}(H)$.

In Refs. [1, 2], Navara and Svozil introduced *pseudocontexts* (or *pseudoblocks*). Under the restrictive affine definition of the latter work, these are disjoint sets $A, B \subseteq V$ of equal size $k \geq 2$, each independent in the two-section graph $\Gamma(H)$, whose probability sums are forced to coincide by a balanced linear combination of context equations. In particular,

$$\sum_{a \in A} p(a) = \sum_{b \in B} p(b) \qquad (\forall\, p \in \mathcal{A}(H)). \tag{4}$$

The common sum need not equal one, and neither set is required to be complete or exclusive under a coloring. When H admits a faithful orthogonal representation (FOR) [3, 4] in $\mathbb{F}^d$, the affine certificate forces the operator identity $\sum_{a \in A} P_a = \sum_{b \in B} P_b$ in every FOR of H. That operator identity imposes geometric rigidity [2, 5]:

1. *No single complete nonorthogonal d-tuple:* If $U = \{u_1, \ldots, u_d\}$ satisfies $\sum_{u \in U} P_u = I_d$ in $\mathbb{C}^d$, then $P_{u_i} P_{u_j} = 0$ for $i \neq j$, so U must be an orthogonal basis rather than a nonorthogonal set.

2. *No size-one ($k = 1$) Hilbert-space pseudocontext:* $P_a = P_b$ implies $[q_a] = [q_b]$, violating ray injectivity.

3. *No size-two ($k = 2$) real Hilbert-space pseudocontext:* In $\mathbb{R}^d$, if $q_{a_1} \not\perp q_{a_2}$, the spectral decomposition of $P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}$ uniquely fixes the unordered pair of rays $\{[q_{a_1}], [q_{a_2}]\} = \{[q_{b_1}], [q_{b_2}]\}$. Thus paired affine pseudocontexts require $k \geq 2$ over $\mathbb{C}$ (attained by a 20-ray octagon H_{20} in $\mathbb{C}^3$) and $k \geq 3$ over $\mathbb{R}$ (attained by Mayet's 15-ray hypergraph $M_{15,8}$ in $\mathbb{R}^3$) [2].

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A parallel line of research studies *common-spectrum spectral assignments* (strong d-colorings) of quantum hypergraphs [6–9]. Assigning each ray $v \in V$ a context-independent eigenvalue $\lambda_{c(v)}$ from a fixed d-element spectrum $\Sigma = \{\lambda_0, \ldots, \lambda_{d-1}\}$ defines nondegenerate commuting observables $A_e = \sum_{v \in e} \lambda_{c(v)} P_v$ [10, 11] if and only if $c : V \to \{0, \ldots, d-1\}$ is a strong d-coloring ($c \in \mathcal{C}_d(H)$), injective on every context $e \in E$. In the companion work [9], strong d-colorings were characterized by $n \times d$ *Context–Color Matrices* $M \in V^{n \times d}$ whose rows permute the d rays of each context and whose d columns are the *extendable* two-valued states $\mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$.

The central question of this paper is different: can a set of pairwise nonadjacent vertices be forced to be complete and exclusive by all global strong d-colorings? We call such a set a *chromatic admissibility context*. Its vertices occupy all d columns of every Context–Color Matrix, exactly once each. More generally, an *admissibility context relative to a nonempty state family* is an independent set whose state sum is identically one on that family. This formulation separates the chromatic case, based on $\mathcal{S}_{\text{ext}}(H)$, from the stronger two-valued case, based on all of $\mathcal{S}(H)$. Neither formulation presupposes an affine or operator identity. Alongside this central notion, we study universal rainbow exclusivity, chromatic twins, and paired color-multiset equivalence. We show that:

- Every balanced affine pseudocontext (A, B) automatically enforces *color-multiset conservation* $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$ in every $n \times d$ Context–Color Matrix (Theorem 3 and Corollary 4).

- Faithfully coordinatized real three-dimensional hypergraphs admit both *chromatic admissibility contexts* (normalized on $\mathcal{S}_{\text{ext}}(H)$, starting at $H_{20,12}$ and $H_{21,13}$) and genuine *two-valued admissibility contexts* (normalized on all of $\mathcal{S}(H)$, realized in $G_{41,25}$ and $G_{41,26}$ via cascaded roof-twin gadgets and in 1–0-extended $H_{20,12}$ and $H_{21,13}$). The examples also exhibit non-twin color-multiset-equivalent pairs of size two, universally rainbow nonorthogonal subsets, and chromatic twin classes, with related phenomena in $\mathbb{R}^4$. Because $\{0,1\}$-integrality on Specker bugs enforces biconditional two-valued twins $s(x) = s(x')$ that do not hold on $\mathcal{A}(H)$ ($\text{aff}\,\mathcal{S}(H) \subsetneq \mathcal{A}(H)$), the operator rigidity statements above remain valid.

Section II defines admissibility contexts and the related chromatic relations through the Context–Color Matrix criterion. Section III analyzes affine pseudocontexts and their color-multiset consequences in $\mathbb{R}^3$ and $\mathbb{C}^3$ ($M_{15,8}$, $B_{13,7}$, $R_{21,12}$, $R_{25,15}$, and H_{20}). Section IV exhibits chromatic admissibility contexts, non-twin color-multiset-equivalent pairs of size two, and genuine two-valued admissibility contexts in real three-dimensional configurations. Section V treats related chromatic relations in $\mathbb{R}^4$, and Section VI summarizes the census and the three-tier distinction among state families.

II. CONTEXT–COLOR MATRICES AND ADMISSIBILITY CONTEXTS

Let $H = (V, E)$ be a finite d-uniform hypergraph with $V = \bigcup_{e \in E} e$ and $n = |E|$ contexts $E = \{C_1, \ldots, C_n\}$. Its two-section graph $\Gamma(H)$ has vertex set V and edges $\{u, v\}$ whenever $u \neq v$ share at least one context $C_i \in E$. A *faithful orthogonal representation* (FOR) in $\mathbb{F}^d$ assigns nonzero vectors $q_v \in \mathbb{F}^d$ to $v \in V$ such that $[q_u] \neq [q_v]$ for $u \neq v$ and $\langle q_u, q_v \rangle = 0$ if and only if $\{u, v\} \in E(\Gamma(H))$. Thus an independent set $U \subseteq V$ in $\Gamma(H)$ is a family of *pairwise nonorthogonal* rays.

By the Universal $n \times d$ Context–Color Matrix Criterion [9, Theorem 1], H admits a strong d-coloring $c \in \mathcal{C}_d(H)$ if and only if the vertices of each context C_i can be ordered into an $n \times d$ matrix $M = (M_{i,j}) \in V^{n \times d}$ ($1 \leq i \leq n$, $0 \leq j \leq d-1$) such that (1) row i is a permutation of C_i, and (2) each vertex $v \in V$ appears in a single column $j = c(v)$. The column vertex sets $S_j = \{M_{i,j} : 1 \leq i \leq n\} = c^{-1}(\{j\})$ form d pairwise disjoint two-valued state supports whose indicator functions $s_j = \mathbf{1}_{S_j} \in \mathcal{S}_{\text{ext}}(H)$ partition V:

$$\sum_{j=0}^{d-1} s_j(v) = 1 \qquad (\forall\, v \in V). \tag{5}$$

In every Context–Color Matrix, each actual context is a complete rainbow row. The following definition isolates the same exclusivity and completeness when they are enforced on vertices that share no context with one another.

Definition 1 (Admissibility Context Relative to a State Family)**.** Let $\varnothing \neq \mathcal{T} \subseteq \mathcal{S}(H)$. An independent set $U \subseteq V$ in $\Gamma(H)$ is an *admissibility context relative to* $\mathcal{T}$ if

$$\sum_{u \in U} s(u) = 1 \qquad (\forall\, s \in \mathcal{T}). \tag{6}$$

The equation enforces both exclusivity (at most one vertex is assigned one) and completeness (at least one is assigned one), despite pairwise nonadjacency. For $\mathcal{T} = \mathcal{S}_{\text{ext}}(H) \neq \varnothing$, we call U a *chromatic admissibility context*; for $\mathcal{T} = \mathcal{S}(H) \neq \varnothing$, we call it a *two-valued admissibility context*. The state family is part of the definition. No affine certificate or projector normalization is assumed.

When $\mathcal{C}_d(H) \neq \varnothing$, every two-valued admissibility context is a chromatic admissibility context because $\mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$. The converse need not hold. The chromatic case has exactly d vertices: for any coloring, sum Eq. (6) over its d color-class states and use Eq. (5) to obtain $|U| = d$. Theorem 3(1) identifies this condition with universal rainbow completeness.

We now distinguish this normalization property from the other universal chromatic relations used below.

Definition 2 (Admissibility and Related Chromatic Relations)**.** Let $H = (V, E)$ be a d-uniform hypergraph with $\mathcal{C}_d(H) \neq \varnothing$.

1. *Class I: Chromatic Admissibility Contexts.* An independent set $U = \{u_1, \dots, u_d\} \subseteq V$ is a chromatic admissibility context precisely when it receives all d colors in every strong d-coloring:

$$\big\{c(u_1), \dots, c(u_d)\big\} = \{0, 1, \dots, d-1\} \qquad \forall\, c \in \mathcal{C}_d(H). \tag{7}$$

If a non-context d-tuple satisfies Eq. (7) but has both adjacent and nonadjacent pairs, we call it a *partially nonorthogonal complete rainbow d-tuple*. It is not an admissibility context under Definition 1, which requires pairwise nonadjacency.

2. *Class II: Universal Rainbow Sets and Chromatic Twin Classes.* Let $U = \{u_1, \dots, u_k\} \subseteq V$ be an independent set with $k \geq 2$.

   (a) U is a *universally rainbow (chromatically exclusive) k-tuple* ($2 \leq k < d$) if its elements receive pairwise distinct colors in every strong d-coloring:

$$c(u_a) \neq c(u_b) \quad (1 \leq a < b \leq k), \qquad \forall\, c \in \mathcal{C}_d(H). \tag{8}$$

   (b) Write $u \sim_c v$ if $c(u) = c(v)$ for every $c \in \mathcal{C}_d(H)$; such vertices are *chromatic twins*. A *chromatic twin class* is an equivalence class of $\sim_c$, and a *universally monochromatic set* is any subset of such a class. Thus U is universally monochromatic (with no upper bound on k) if

$$c(u_1) = c(u_2) = \cdots = c(u_k) \qquad \forall\, c \in \mathcal{C}_d(H). \tag{9}$$

3. *Class III: Color-Multiset-Equivalent Pairs.* Two disjoint independent sets $A = \{a_1, \dots, a_k\}$ and $B = \{b_1, \dots, b_k\}$, with $2 \leq k \leq d$, form a *color-multiset-equivalent pair* if

$$\big\{\{c(a) : a \in A\}\big\} = \big\{\{c(b) : b \in B\}\big\} \qquad \forall\, c \in \mathcal{C}_d(H). \tag{10}$$

The pair is *non-twin* if no element of A is a chromatic twin of an element of B ($\neg(a_i \sim_c b_j)$ for all $a_i \in A$, $b_j \in B$). Color-multiset equivalence alone does not impose exclusivity or completeness on either side. We reserve *affine pseudocontext* for a pair with a balanced context certificate. In the census, we count non-twin pairs and omit pairs in which both sides are already chromatic admissibility contexts; their equality follows directly from the single-set normalizations.

Theorem 3 (Chromatic Relations via Extendable State Sums)**.** *Let $H = (V, E)$ be a d-uniform hypergraph with $\mathcal{C}_d(H) \neq \varnothing$ and extendable state set $\mathcal{S}_{\text{ext}}(H) = \{s_j = \mathbf{1}_{c^{-1}(\{j\})} : c \in \mathcal{C}_d(H),\ 0 \leq j \leq d-1\}$. Then:*

1. *A d-element set $U = \{u_1, \dots, u_d\} \subseteq V$ satisfies $\{c(u) : u \in U\} = \{0, \dots, d-1\}$ for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{u \in U} s(u) = 1 \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{11}$$

2. *A k-element set $U = \{u_1, \dots, u_k\} \subseteq V$ ($2 \leq k < d$) satisfies $c(u_a) \neq c(u_b)$ ($a \neq b$) for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{u \in U} s(u) \leq 1 \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{12}$$

3. *Disjoint k-element sets $A, B \subseteq V$ satisfy $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$ for all $c \in \mathcal{C}_d(H)$ if and only if*

$$\sum_{a \in A} s(a) = \sum_{b \in B} s(b) \qquad \forall\, s \in \mathcal{S}_{\text{ext}}(H). \tag{13}$$

Proof. For any $c \in \mathcal{C}_d(H)$, let $s_j = \mathbf{1}_{c^{-1}(\{j\})} \in \mathcal{S}_{\text{ext}}(H)$ ($0 \leq j \leq d-1$) be the indicator functions of the d columns of the associated $n \times d$ Context–Color Matrix M. For any subset $X \subseteq V$ and color $j \in \{0, \dots, d-1\}$, the column sum

$$\sum_{x \in X} s_j(x) = \big|\{x \in X : c(x) = j\}\big| \tag{14}$$

counts the exact multiplicity of color j on X in coloring c. (1) When $|U| = d$, every color $j \in \{0, \dots, d-1\}$ has multiplicity 1 on U iff $\{c(u) : u \in U\} = \{0, \dots, d-1\}$, establishing Eq. (11). (2) Every color j has multiplicity at most 1 on U iff no two elements of U share a color in c, establishing Eq. (12). (3) By Eq. (14), color j has the same multiplicity on A and on B for every $j \in \{0, \dots, d-1\}$ and every $c \in \mathcal{C}_d(H)$ iff $\sum_{a \in A} s_j(a) = \sum_{b \in B} s_j(b)$ for every $s_j \in \mathcal{S}_{\text{ext}}(H)$, establishing Eq. (13). □

Corollary 4 (Affine Pseudocontexts Imply Color-Multiset Equivalence)**.** *Let $M_H \in \{0,1\}^{n \times |V|}$ be the context-vertex incidence matrix of H. If disjoint nonorthogonal k-tuples $A, B \subseteq V$ admit a balanced context certificate $\lambda \in \mathbb{R}^n$ satisfying [2]*

$$M_H^{\mathsf{T}} \lambda = \chi_A - \chi_B, \qquad \mathbf{1}_n^{\mathsf{T}} \lambda = 0, \tag{15}$$

then in every strong d-coloring $c \in \mathcal{C}_d(H)$, A and B carry the exact same multiset of colors: $\{\{c(a) : a \in A\}\} = \{\{c(b) : b \in B\}\}$.

Proof. Every extendable state $s \in \mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$ satisfies $M_H s = \mathbf{1}_n$, so

$$\sum_{a \in A} s(a) - \sum_{b \in B} s(b) = (\chi_A - \chi_B)^{\mathsf{T}} s = \lambda^{\mathsf{T}} M_H s = \lambda^{\mathsf{T}} \mathbf{1}_n = 0.$$

Applying Theorem 3(3) completes the proof. □

III. PAIRED AFFINE-FORCED PSEUDOCONTEXTS IN $\mathbb{R}^3$ AND $\mathbb{C}^3$

A. Mayet's 15-Ray, 8-Context Real Configuration $M_{15,8}$

Consider the 15-vertex, 8-context hypergraph $M_{15,8}$ studied in Refs. [1, 2, 12, 13], with contexts

$$\begin{aligned} C_1 &= \{1,2,3\}, & C_2 &= \{3,4,5\}, \\ C_3 &= \{6,7,8\}, & C_4 &= \{8,9,10\}, \\ C_5 &= \{11,12,13\}, & C_6 &= \{13,14,15\}, \\ C_7 &= \{2,7,12\}, & C_8 &= \{4,9,14\}. \end{aligned} \tag{16}$$

Clearing denominators in Eqs. (9) of Ref. [1] supplies an exact integer FOR in $\mathbb{R}^3$ (zero extra orthogonalities among all $\binom{15}{2} = 105$ pairs):

$$\begin{aligned} r_1 &= (5,3,-4), & r_2 &= (5,-7,1), \\ r_3 &= (1,1,2), & r_4 &= (1,1,-1), \\ r_5 &= (1,-1,0), & r_6 &= (5,-16,13), \\ r_7 &= (5,4,3), & r_8 &= (-2,1,2), \\ r_9 &= (1,0,1), & r_{10} &= (1,4,-1), \\ r_{11} &= (-5,18,1), & r_{12} &= (5,2,-11), \\ r_{13} &= (4,1,2), & r_{14} &= (-1,2,1), \\ r_{15} &= (1,2,-3). \end{aligned} \tag{17}$$

The two primary target triples $A = \{1,6,11\}$ and $B = \{5,10,15\}$ are internally pairwise nonorthogonal. The two 4-context coverings $\mathcal{F}_+ = \{C_1, C_3, C_5, C_8\}$ and $\mathcal{F}_- = \{C_2, C_4, C_6, C_7\}$ partition $V \setminus B$ and $V \setminus A$, respectively, giving $\sum_{C\in\mathcal{F}_+} \chi_C - \sum_{C\in\mathcal{F}_-} \chi_C = \chi_A - \chi_B$. In addition, omitting any one of the three horizontal 2-context chains (C_1, C_2), (C_3, C_4), or (C_5, C_6) leaves a 6-context hexagon whose 3-context coverings force three further paired size-three affine pseudocontexts: $\{1,6,14\}$ vs. $\{5,10,12\}$, $\{1,9,11\}$ vs. $\{5,7,15\}$, and $\{4,6,11\}$ vs. $\{2,10,15\}$ (4 pairs in total).

Exhaustive enumeration gives 24 two-valued states of $M_{15,8}$, all extendable, and $N_c = 16$ normalized 8×3 Context–Color Matrices (with anchor $C_1 = (1,2,3)$). By Corollary 4, the two target triples have identical color multisets in every coloring:

- In 4 normalized colorings, both A and B are *rainbow* $\{0,1,2\}$.
- In 12 normalized colorings, both A and B carry a *two-color multiset*: 4 with $\{0,0,1\}$, 4 with $\{0,1,1\}$, and 4 with $\{0,0,2\}$. (Neither triple is ever monochromatic because $p(1)+p(6)+p(11) \le 2$ on $\mathcal{S}(M_{15,8})$ [1].)

Two explicit 8×3 Context–Color Matrices illustrating the rainbow regime ($M_1^{(M_{15,8})}$) and the two-color multiset regime ($M_2^{(M_{15,8})}$) are:

$$M_1^{(M_{15,8})} = \left(\begin{array}{ccc} \mathbf{c_0} & \mathbf{c_1} & \mathbf{c_2} \\ \hline \mathbf{1} & 2 & 3 \\ 4 & \mathbf{5} & 3 \\ 8 & \mathbf{6} & 7 \\ 8 & 9 & \mathbf{10} \\ 12 & 13 & \mathbf{11} \\ \mathbf{15} & 13 & 14 \\ \hline 12 & 2 & 7 \\ 4 & 9 & 14 \end{array}\right), \quad M_2^{(M_{15,8})} = \left(\begin{array}{ccc} \mathbf{c_0} & \mathbf{c_1} & \mathbf{c_2} \\ \hline \mathbf{1} & 2 & 3 \\ 4 & \mathbf{5} & 3 \\ \mathbf{6} & 8 & 7 \\ \mathbf{10} & 8 & 9 \\ 12 & \mathbf{11} & 13 \\ \mathbf{15} & 14 & 13 \\ \hline 12 & 2 & 7 \\ 4 & 14 & 9 \end{array}\right). \tag{18}$$

In $M_1^{(M_{15,8})}$, both $A = \{1,6,11\}$ and $B = \{5,10,15\}$ span all three columns $(\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2)$ via $(1,6,11)$ and $(15,5,10)$. In $M_2^{(M_{15,8})}$, both A and B have two entries in Column $\mathbf{c}_0$ ($\{1,6\}$ and $\{10,15\}$), one entry in Column $\mathbf{c}_1$ (11 and 5), and none in Column $\mathbf{c}_2$. Figure 1 displays both colorings on the Greechie diagram of $M_{15,8}$.

B. Balanced Size-Three Pseudocontexts in $B_{13,7}$ and $R_{21,12}, R_{25,15}$

Two further families of faithfully coordinatized hypergraphs from Ref. [9] already contain balanced covering certificates for size-three affine pseudocontexts:

1. *The 13-ray, 7-context Specker bug $B_{13,7} \subset \mathbb{R}^3$ ($N_c = 6$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 12/14$):* Its 7 contexts are $E_1 = \{1,2,3\}$, $E_2 = \{7,8,9\}$, $E_3 = \{2,8,5\}$, $E_4 = \{3,11,14\}$, $E_5 = \{7,11,15\}$, $E_6 = \{9,10,23\}$, and $E_7 = \{1,10,19\}$ [Fig. 3(a)], with Yu–Oh coordinates from Table I. The six outer contexts partition into two 3-context coverings $\mathcal{F}_+ = \{E_1, E_5, E_6\}$ and $\mathcal{F}_- = \{E_2, E_4, E_7\}$ of $(V \setminus \{5\}) \setminus B$ and $(V \setminus \{5\}) \setminus A$, where

$$A = \{2,15,23\}, \qquad B = \{8,14,19\}. \tag{19}$$

   Hence $(\chi_{E_1} + \chi_{E_5} + \chi_{E_6}) - (\chi_{E_2} + \chi_{E_4} + \chi_{E_7}) = \chi_A - \chi_B$, forcing $\{\{c(2), c(15), c(23)\}\} = \{\{c(8), c(14), c(19)\}\}$ across all $N_c = 6$ normalized 7×3 Context–Color Matrices (and likewise in the 4-dimensional common-apex lift $B_{14,7}^{(4)}$).

2. *Gaussian-integer row–column hub grids $R_{21,12}$ and $R_{25,15}$ in $\mathbb{C}^3$ [9, Sec. IV D]:* In $R_{21,12}$ ($N_c = 32$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 48/54$) and $R_{25,15}$ ($N_c = 24$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 48/81$), the 3×3 core grid has row contexts $R_1 = \{1,2,3\}$, $R_2 = \{4,5,6\}$, $R_3 = \{7,8,9\}$ and hub contexts $\{1,10,13\}$, $\{2,11,14\}$, $\{3,12,15\}$ (for R_1), $\{4,10,16\}$, $\{5,11,17\}$, $\{6,12,18\}$ (for R_2), and $\{7,10,19\}$, $\{8,11,20\}$, $\{9,12,21\}$ (for R_3). For any two rows (say R_1 and R_2) and any column $j \in \{1,2,3\}$ (say $j = 1$), the two 3-context families $\mathcal{F}_+ = \{\{1,2,3\}, \{4,10,16\}, \{2,11,14\}, \{3,12,15\}\} \setminus$

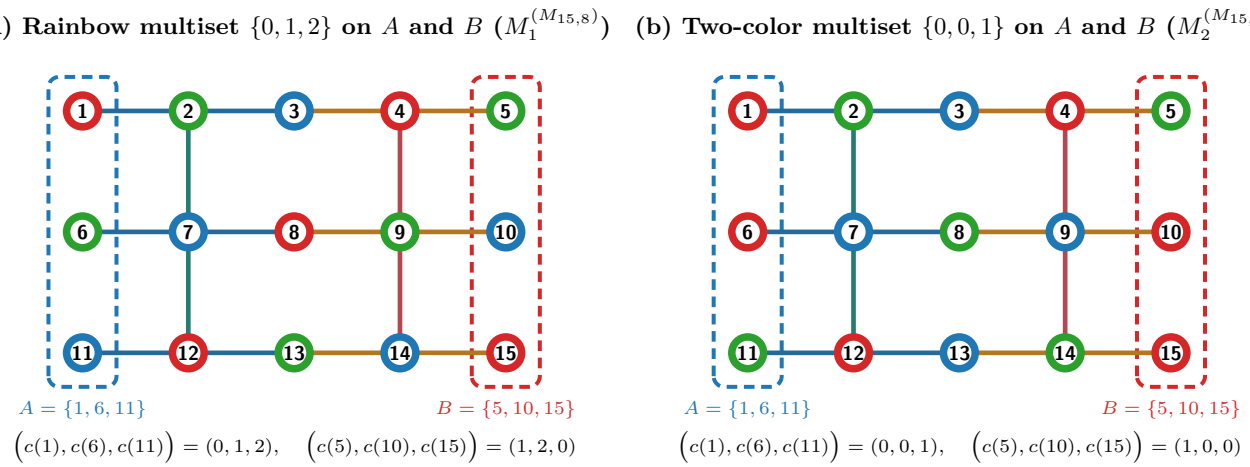


FIG. 1. Mayet's 15-ray, 8-context hypergraph $M_{15,8}$ in $\mathbb{R}^3$ [Eq. (17)] with paired size-three affine pseudocontexts $A = \{1,6,11\}$ and $B = \{5,10,15\}$ (dashed boxes). Vertex outer rings indicate colors 0 (red), 1 (green), and 2 (blue) for the two Context–Color Matrices in Eq. (18): (a) $M_1^{(M_{15,8})}$, where both A and B are rainbow $\{0,1,2\}$; (b) $M_2^{(M_{15,8})}$, where both A and B carry the two-color multiset $\{0,0,1\}$.

$\{\{1,2,3\}\}$ combine to yield 12 size-three affine pseudocontexts such as

$$A = \{1,17,18\}, \qquad B = \{4,14,15\}, \tag{20}$$

via the balanced 3-context identity $(\chi_{\{1,2,3\}} + \chi_{\{5,11,17\}} + \chi_{\{6,12,18\}}) - (\chi_{\{4,5,6\}} + \chi_{\{2,11,14\}} + \chi_{\{3,12,15\}}) = \chi_{\{1,17,18\}} - \chi_{\{4,14,15\}}$, forcing $\{\{c(1), c(17), c(18)\}\} = \{\{c(4), c(14), c(15)\}\}$ in all 32 (resp. 24) normalized Context–Color Matrices.

C. The 20-Ray, 12-Context Complex Octagon $\mathsf{H}_{20} \subset \mathbb{C}^3$

To reach target size $k = 2$ via a balanced covering certificate, one must pass from $\mathbb{R}^3$ to $\mathbb{C}^3$ [2, 5]. The 20-ray, 12-context octagon H_{20} (Fig. 2) has eight perimeter contexts $E_j = \{v_{j-1,j}, v_j, v_{j,j+1}\}$ ($j \in \mathbb{Z}_8$) and four inner contexts $D_0 = \{v_0, a_1, v_4\}$, $D_2 = \{v_2, a_2, v_6\}$, $D_1 = \{v_1, b_2, v_5\}$, $D_3 = \{v_3, b_1, v_7\}$. The two 6-context coverings $\mathcal{G}_+ = \{D_0, D_2, E_1, E_3, E_5, E_7\}$ and $\mathcal{G}_- = \{D_1, D_3, E_0, E_2, E_4, E_6\}$ yield $\sum_{C\in\mathcal{G}_+} \chi_C - \sum_{C\in\mathcal{G}_-} \chi_C = \chi_A - \chi_B$ for the nonorthogonal pairs

$$A = \{a_1, a_2\}, \qquad B = \{b_1, b_2\}. \tag{21}$$

Exhaustive computation gives $|\mathcal{S}(\mathsf{H}_{20})| = |\mathcal{S}_{\text{ext}}(\mathsf{H}_{20})| = 26$ two-valued states and $N_c = 12$ normalized 12×3 Context–Color Matrices (with anchor $E_0 = (v_{70}, v_0, v_{01})$). Across all 12 normalized colorings, $\{\{c(a_1), c(a_2)\}\} = \{\{c(b_1), c(b_2)\}\}$ holds identically: 4 colorings assign monochromatic multisets (2 with $\{0,0\}$ and 2 with $\{2,2\}$) and 8 assign bichromatic multisets (2 with $\{0,1\}$, 4 with $\{0,2\}$, and 2 with $\{1,2\}$). Representative bichromatic ($M_1^{(\mathsf{H}_{20})}$) and monochromatic ($M_2^{(\mathsf{H}_{20})}$) Context–Color Matrices are:

$$M_1^{(\mathsf{H}_{20})} = \left(\begin{array}{ccc} \mathbf{c_0} & \mathbf{c_1} & \mathbf{c_2} \\ \hline v_{70} & v_0 & v_{01} \\ v_1 & v_{12} & v_{01} \\ v_{23} & v_{12} & v_2 \\ v_{23} & v_{34} & v_3 \\ v_4 & v_{34} & v_{45} \\ v_{56} & v_5 & v_{45} \\ v_{56} & v_6 & v_{67} \\ v_{70} & v_7 & v_{67} \\ \hline v_4 & v_0 & \mathbf{a_1} \\ \mathbf{a_2} & v_6 & v_2 \\ v_1 & v_5 & \mathbf{b_2} \\ \mathbf{b_1} & v_7 & v_3 \end{array}\right), \quad M_2^{(\mathsf{H}_{20})} = \left(\begin{array}{ccc} \mathbf{c_0} & \mathbf{c_1} & \mathbf{c_2} \\ \hline v_{70} & v_0 & v_{01} \\ v_{12} & v_1 & v_{01} \\ v_{12} & v_2 & v_{23} \\ v_{34} & v_3 & v_{23} \\ v_{34} & v_{45} & v_4 \\ v_{56} & v_{45} & v_5 \\ v_{56} & v_{67} & v_6 \\ v_{70} & v_{67} & v_7 \\ \hline \mathbf{a_1} & v_0 & v_4 \\ \mathbf{a_2} & v_2 & v_6 \\ \mathbf{b_2} & v_1 & v_5 \\ \mathbf{b_1} & v_3 & v_7 \end{array}\right). \tag{22}$$

IV. CHROMATIC ADMISSIBILITY CONTEXTS AND COLOR-MULTISET EQUIVALENCE IN $\mathbb{R}^3$

A. Chromatic Admissibility and Operator Rigidity

Before exhibiting chromatic admissibility contexts in $\mathbb{R}^3$, we distinguish their state-restricted normalization from an affine or operator identity.

Proposition 5 (Operator and Affine Rigidity of Complete d-Tuples)**.** *Let $H = (V, E)$ be a d-uniform hypergraph admitting a FOR $v \mapsto q_v \in \mathbb{C}^d$ with rank-one projectors $P_v = q_v q_v^\dagger / \|q_v\|^2$. If a d-element subset $U = \{u_1, \ldots, u_d\} \subseteq V$ satisfies $\sum_{u\in U} P_u = I_d$ (in particular, if $\chi_U = M_H^{\mathsf{T}} \lambda$ with $\mathbf{1}_n^{\mathsf{T}} \lambda = 1$), then $q_{u_a} \perp q_{u_b}$ for all $1 \le a < b \le d$. Consequently, no set of d nonorthogonal*

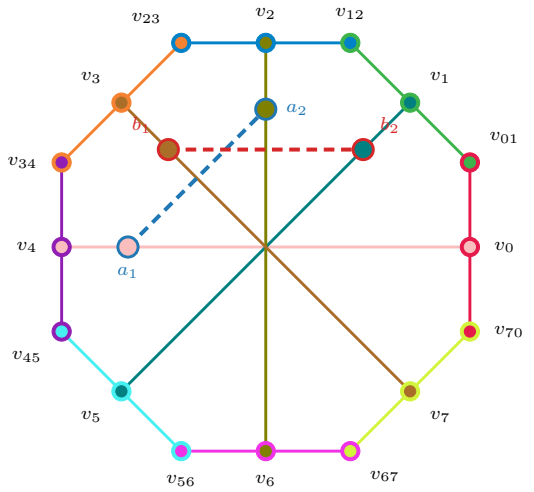


FIG. 2. The 20-ray, 12-context complex octagon $\mathsf{H}_{20} \subset \mathbb{C}^3$ [2, 5]. The two nonorthogonal target pairs $A = \{a_1, a_2\}$ (blue dashed link) and $B = \{b_1, b_2\}$ (red dashed link) carry identical 2-element color multisets $\{\{c(a_1), c(a_2)\}\} = \{\{c(b_1), c(b_2)\}\}$ across all $N_c = 12$ normalized 12×3 Context–Color Matrices [Eq. (22)].

rays can sum to I_d or be forced to sum to 1 *by a linear combination of context equations.*

Proof. For each $a \in \{1, \ldots, d\}$, conjugating $I_d - P_{u_a} = \sum_{b \neq a} P_{u_b}$ by the unit vector $\hat{q}_{u_a} = q_{u_a}/\|q_{u_a}\|$ gives

$$0 = \langle \hat{q}_{u_a}, (I_d - P_{u_a})\hat{q}_{u_a} \rangle = \sum_{b \neq a} \left| \langle \hat{q}_{u_a}, \hat{q}_{u_b} \rangle \right|^2,$$

so $\langle q_{u_a}, q_{u_b} \rangle = 0$ for all $b \neq a$. If $\chi_U = M_H^{\mathsf{T}} \lambda$ with $\mathbf{1}_n^{\mathsf{T}} \lambda = 1$, summing $\sum_{v \in C_i} P_v = I_d$ with weights λ_i yields $\sum_{u \in U} P_u = (\mathbf{1}_n^{\mathsf{T}} \lambda) I_d = I_d$, forcing mutual orthogonality. □

Proposition 5 excludes operator completeness and an affine normalization certificate for any d-tuple containing a nonorthogonal pair. It does not exclude normalization on a specified family of two-valued states. In particular, chromatic admissibility imposes Eq. (11) only on $\mathcal{S}_{\text{ext}}(H)$. Whether this identity extends to all of $\mathcal{S}(H)$ is a separate question; the proposition alone does not imply that a violating nonextendable state must exist.

A sufficient additional hypothesis for lifting every identity on two-valued states to an affine identity is

$$\operatorname{aff} \mathcal{S}(H) = \mathcal{A}(H). \tag{23}$$

Under this hypothesis, normalization on all of $\mathcal{S}(H)$ extends to $\mathcal{A}(H)$ and hence has a context-equation certificate. Consequently, in a faithful orthogonal representation, an independent chromatic admissibility context must then violate normalization on some nonextendable two-valued state. Without Eq. (23), no such general necessity is asserted. The examples below distinguish the two state families directly through their enumerated state sums.

B. Non-Twin Color-Multiset-Equivalent Pairs of Size Two in $\mathbb{R}^3$ ($I_{16,9}, H_{17,10}, T_{19,11}$)

A faithful real three-dimensional hypergraph can support a non-twin color-multiset-equivalent pair of size $k = 2$ without an affine certificate. This does not contradict the prohibition of size-two real affine pseudocontexts in Ref. [2]: the color relation is required only on $\mathcal{S}_{\text{ext}}(H)$. The construction combines a linear context identity with a non-context complete rainbow triple.

Theorem 6 (Non-Twin Color-Multiset Equivalence of Size Two in $\mathbb{R}^3$)**.** *In the faithful real three-dimensional hypergraphs $I_{16,9}$ (16 rays, 9 contexts, $N_c = 6$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 12/22$) and $H_{17,10}$ (17 rays, 10 contexts, $N_c = 4$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 10/18$), the nonorthogonal pairs*

$$\begin{aligned} A_1 &= \{2, 4\}, \quad B_1 = \{14, 19\}, \\ A_2 &= \{4, 8\}, \quad B_2 = \{15, 23\} \end{aligned} \tag{24}$$

have no chromatic twins between A_r and B_r, yet satisfy

$$\begin{aligned} \{\{c(2), c(4)\}\} &= \{\{c(14), c(19)\}\}, \\ \{\{c(4), c(8)\}\} &= \{\{c(15), c(23)\}\} \end{aligned} \qquad \forall\, c \in \mathcal{C}_3. \tag{25}$$

Similarly, in $T_{19,11}$ (19 rays, 11 contexts, $N_c = 7$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 14/25$, Fig. 3(b)), the partially nonorthogonal tip triple $U_{\text{tip}} = \{26, 27, 28\}$ (with $26 \perp 27$, $26 \perp 28$, $27 \not\perp 28$) is rainbow in every 3-coloring, and the nonorthogonal pairs $A_3 = \{3, 10\}$, $B_3 = \{24, 26\}$ satisfy $\{\{c(3), c(10)\}\} = \{\{c(24), c(26)\}\}$ for all $c \in \mathcal{C}_3(T_{19,11})$.

Proof. Both $I_{16,9}$ and $H_{17,10}$ contain the Specker-bug contexts $E_1 = \{1, 2, 3\}$, $E_2 = \{7, 8, 9\}$, $E_4 = \{2, 8, 5\}$, $E_8 = \{3, 11, 14\}$, $E_9 = \{7, 11, 15\}$, $E_{11} = \{9, 10, 23\}$, $E_{12} = \{1, 10, 19\}$ together with the two roof contexts $E_{10} = \{4, 11, 24\}$ and $E_{13} = \{4, 10, 18\}$ ($H_{17,10}$ also includes $E_3 = \{4, 5, 6\}$). On the Specker bug, every 3-coloring satisfies the universally rainbow condition $c(10) \neq c(11)$. Since 4 is adjacent to 11 in E_{10} and to 10 in E_{13}, $c(4) \neq c(11)$ and $c(4) \neq c(10)$. Thus $\{4, 10, 11\}$ is a partially nonorthogonal complete rainbow triple: $\{c(4), c(10), c(11)\} = \{0, 1, 2\}$ for all $c \in \mathcal{C}_3$, or equivalently by Theorem 3(1),

$$s(4) + s(10) + s(11) = 1 \qquad \forall\, s \in \mathcal{S}_{\text{ext}}. \tag{26}$$

Meanwhile, on *every* two-valued state $s \in \mathcal{S}$, the context combination $E_8 + E_{12} - E_1$ gives

$$\begin{aligned} \big(s(3) + s(11) + s(14)\big) + \big(s(1) + s(10) + s(19)\big) \\ - \big(s(1) + s(2) + s(3)\big) = 1, \end{aligned} \tag{27}$$

so $s(14) + s(19) - s(2) = 1 - s(10) - s(11)$. Substituting Eq. (26) into Eq. (27) yields $s(2) + s(4) = s(14) + s(19)$ for all $s \in \mathcal{S}_{\text{ext}}$, and symmetric combination $E_9 + E_{11} - E_2$ yields $s(4) + s(8) = s(15) + s(23)$ for all $s \in \mathcal{S}_{\text{ext}}$. By Theorem 3(3), Eq. (25) holds in every 3-coloring. In $H_{17,10}$ [Eq. (14) of Ref. [9]], $(c(2), c(4))$ and $(c(14), c(19))$

TABLE I. Unnormalized integer vectors in $\mathbb{R}^3$ for the 25-ray completed Yu–Oh configuration H^+_{YO} (rays $1,\dots,25$, used by $B_{13,7}$, $H_{17,10}$, and $H_{21,13}$) and for the 19-ray gadget $T_{19,11}$ (second tuple where different, plus tip rays $26,27,28$) [8, 14]. For $H_{20,12}$ (omitting $\{6,13,22\}$), taking $r_1=(0,1,2)$, $r_3=(0,2,-1)$, $r_7=(2,0,1)$, $r_9=(1,0,-2)$, $r_{10}=(2,-2,1)$, $r_{11}=(1,-1,-2)$, $r_{13}=(1,4,-2)$ and cross-product completions yields a strictly faithful integer representation with $\langle r_6, r_{13}\rangle = -3 \neq 0$.

i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)	i	r_i (H^+_{YO} / $T_{19,11}$)
1	$(0,1,1)$ / $(0,1,2)$	8	$(0,1,0)$	15	$(1,2,-1)$ / $(1,4,-1)$	22	$(1,1,2)$
2	$(1,0,0)$	9	$(1,0,-1)$	16	$(1,-2,1)$	23	$(1,2,1)$ / $(1,4,1)$
3	$(0,1,-1)$ / $(0,2,-1)$	10	$(1,-1,1)$ / $(2,-1,2)$	17	$(1,1,-2)$	24	$(1,-1,2)$ / $(4,-2,5)$
4	$(1,1,0)$ / $(1,2,0)$	11	$(1,-1,-1)$ / $(2,-1,-2)$	18	$(1,-1,-2)$ / $(4,-2,-5)$	25	$(2,-1,-1)$
5	$(0,0,1)$	12	$(1,1,1)$	19	$(2,1,-1)$	26	$(2,-41,18)$ ($T_{19,11}$)
6	$(1,-1,0)$ / $(2,-1,0)$	13	$(1,1,-1)$	20	$(2,-1,1)$	27	$(200,-82,-209)$ ($T_{19,11}$)
7	$(1,0,1)$	14	$(2,1,1)$ / $(5,2,4)$	21	$(1,-2,-1)$	28	$(241,82,160)$ ($T_{19,11}$)

equal $(0,0)$ and $(0,0)$ in c_1,c_3, $(0,1)$ and $(1,0)$ in c_2, and $(0,1)$ and $(0,1)$ in c_4, confirming the absence of chromatic twins between A_1 and B_1.

In $T_{19,11}$ [Fig. 3(b)], the two upper contexts $\{14,27,26\}$ and $\{18,28,26\}$ together with the universally rainbow pair $c(14)\neq c(18)$ force $18\sim_c 27$, $14\sim_c 28$, and $\{c(26),c(27),c(28)\}=\{0,1,2\}$. Combining $\{3,11,14\}+\{4,10,18\}-\{4,11,24\}$ with $c(18)=c(27)$ and $\{14,27,26\}$ gives $s(3)+s(10)=s(24)+s(26)$ on $\mathcal{S}_{\mathrm{ext}}(T_{19,11})$. □

C. Chromatic Admissibility Contexts in $H_{20,12}$ and $H_{21,13}$

Finally, we turn to Class I of Definition 2: a single triple $U=\{u_1,u_2,u_3\}$ of *pairwise nonorthogonal* rays in $\mathbb{R}^3$ ($q_{u_a}\not\perp q_{u_b}$ for all $a\neq b$) that is chromatically complete ($\{c(u_1),c(u_2),c(u_3)\}=\{0,1,2\}$) in *every* strong 3-coloring $c\in\mathcal{C}_3(H)$.

Theorem 7 (Sharp Minimality and Transversal Classification in Yu–Oh Restrictions)**.** *Consider the* 13*-context,* 21*-ray Yu–Oh restriction* $H_{21,13}$ *in* $\mathbb{R}^3$ *(Table I and Fig. 4) obtained by omitting the single diagonal sector* $h_0=12$ *from* H^+_{YO} *[8, 9].*

1. Sharp threshold at 12 contexts ($H_{20,12}$): *Among all* $2^{13}=8{,}192$ *context subhypergraphs of* $H_{21,13}$*, none with* $|E|\leq 11$ *contexts possesses a pairwise-nonorthogonal complete rainbow triple. At* $|E|=12$*, deleting a single roof context from one diagonal sector—for instance omitting* $\{6,13,22\}$ *to form* $H_{20,12}$ *(*$|V|=20$*,* $|E|=12$*,* $N_c=4$*,* $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}|=10/25$*)—yields the minimal configuration, which already forces two pairwise-nonorthogonal complete rainbow triples:*

$$U_1^{(20)}=\{15,18,20\},\qquad U_2^{(20)}=\{19,21,24\}. \tag{28}$$

2. Even-transversal classification in $H_{21,13}$ ($|V|=21$, $|E|=13$, $N_c=2$, $|\mathcal{S}_{\mathrm{ext}}|/|\mathcal{S}|=6/21$): *Each vertex* $v\in V(H_{21,13})$ *has a bivariate color signature* $\sigma(v)=(c_1(v),c_2(v))\in\{0,1,2\}^2$ *across the two normalized* 13×3 *Context–Color Matrices [Eq. (16) of Ref. [9]], partitioning* $V(H_{21,13})$ *into the* 3×3 *signature grid:*

$c_2\backslash c_1$	$\mathbf{c}_0=0$	$\mathbf{c}_1=1$	$\mathbf{c}_2=2$
0	$\{2,14\}$	$\{6\}$	$\{7,10,20,24\}$
1	$\{4,13,15,19\}$	$\{8,23\}$	$\{3\}$
2	$\{9\}$	$\{1,11,18,21\}$	$\{5,22\}$

(29)

A triple $U\subseteq V(H_{21,13})$ *is rainbow in every* 3*-coloring if and only if* U *is a* permutation transversal *[15] of Eq. (29), selecting one vertex from each row and one from each column (*$\sum_{\pi\in S_3}\prod_{i=0}^{2}|F_{i,\pi(i)}|=97$ *total transversals, i.e.,* 13 *contexts plus* 84 *non-context rainbow triples). The three odd permutations* $\pi\in S_3$ *contain* 0 *pairwise-nonorthogonal triples, whereas the three even (cyclic) permutations yield* exactly 21 chromatic admissibility contexts*:*

- Singleton cyclic transversal $F_{0,2}\times F_{1,0}\times F_{2,1}$ (1 triple):

$$U_{\mathrm{cop}}=\{3,6,9\}, \tag{30}$$

  whose rays $r_3=(0,1,-1)$*,* $r_6=(1,-1,0)$*,* $r_9=(1,0,-1)$ *are not only pairwise nonorthogonal (*$\langle r_3,r_6\rangle=-1$*,* $\langle r_3,r_9\rangle=1$*,* $\langle r_6,r_9\rangle=1$*) but* coplanar *(*$r_3+r_6-r_9=\mathbf{0}$*, spanning the plane orthogonal to omitted hub* $r_{12}=(1,1,1)$ *so* $\mathrm{rank}(P_3+P_6+P_9)=2<3$*).*

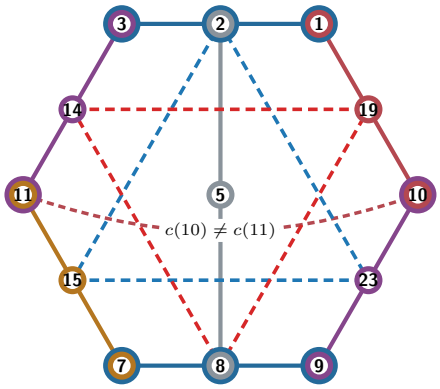
- Main diagonal transversal $F_{0,0}\times F_{1,1}\times F_{2,2}$ (4 of 8 triples):

$$\{2,22,23\},\quad \{5,14,23\},\quad \{8,14,22\},\quad \{14,22,23\}. \tag{31}$$

- Cyclic transversal $F_{0,1}\times F_{1,2}\times F_{2,0}$ (16 of 64 triples):

$$\begin{aligned}&\{1,4,7\},\ \{1,15,24\},\ \{4,20,21\},\ \{7,18,19\},\\ &\{10,11,13\},\ \{10,15,21\},\ \{11,19,20\},\ \{13,18,24\},\\ &\{15,18,20\},\ \{15,18,24\},\ \{15,20,21\},\ \{15,21,24\},\\ &\{18,19,20\},\ \{18,19,24\},\ \{19,20,21\},\ \{19,21,24\}.\end{aligned} \tag{32}$$

(a) Specker bug $B_{13,7}$: $c(10) \neq c(11)$ and $A = \{2, 15, 23\}$, $B = \{8, 14, 19\}$



(b) $T_{19,11}$: rainbow $\{26, 27, 28\}$ and size-two $A_3 = \{3, 10\}$, $B_3 = \{24, 26\}$

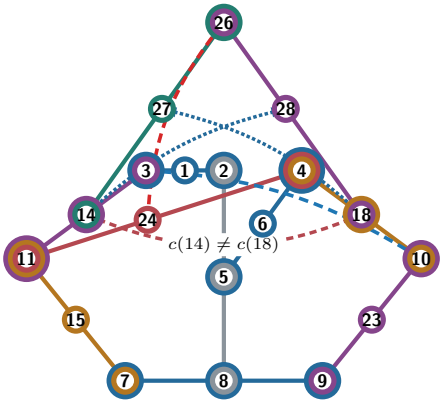


FIG. 3. Three-dimensional gadgets in $\mathbb{R}^3$ (coordinates in Table I): (a) Specker bug $B_{13,7}$ with universally rainbow pair $c(10) \neq c(11)$ and size-three affine pseudocontexts $A = \{2, 15, 23\}$ (blue dashed triangle) and $B = \{8, 14, 19\}$ (red dashed triangle); (b) $T_{19,11}$ with universally rainbow pair $c(14) \neq c(18)$, chromatic twins $14 \sim_c 28$ and $18 \sim_c 27$, complete rainbow tip triple $\{26, 27, 28\}$, and the non-twin color-multiset-equivalent pair $A_3 = \{3, 10\}$ (blue dashed arc) and $B_3 = \{24, 26\}$ (red dashed arc).

In particular, $U_{\text{core}} = \{1, 4, 7\}$ *consists of three core rays* $r_1 = (0, 1, 1)$, $r_4 = (1, 1, 0)$, $r_7 = (1, 0, 1)$ *at pairwise* 60°, *and* $U_{\text{hub}} = \{10, 11, 13\}$ *consists of the three retained diagonal hubs* $r_{10} = (1, -1, 1)$, $r_{11} = (1, -1, -1)$, $r_{13} = (1, 1, -1)$.

3. Chromatic twin classes of size four ($k = 4 > d = 3$): *All four rays within each of the three 4-element signature fibers* $F_{0,2} = \{7, 10, 20, 24\}$, $F_{1,0} = \{4, 13, 15, 19\}$, *and* $F_{2,1} = \{1, 11, 18, 21\}$ *of Eq. (29) are pairwise nonorthogonal in* $\mathbb{R}^3$ *and share the exact same color in every 3-coloring of* $H_{21,13}$.

Proof. In $H_{21,13}$, any strong 3-coloring $c \in \mathcal{C}_3(H_{21,13})$ is obtained by applying a color permutation $\tau \in S_3$ to either c_1 or c_2. Hence $\{c(u_1), c(u_2), c(u_3)\} = \{0, 1, 2\}$ holds for all $c \in \mathcal{C}_3(H_{21,13})$ iff $\{c_1(u_1), c_1(u_2), c_1(u_3)\} = \{0, 1, 2\}$ and $\{c_2(u_1), c_2(u_2), c_2(u_3)\} = \{0, 1, 2\}$, meaning that the three signatures $\sigma(u_1), \sigma(u_2), \sigma(u_3)$ have distinct first coordinates and distinct second coordinates—the definition of a permutation transversal of Eq. (29). Checking inner products $\langle r_u, r_v \rangle \neq 0$ from Table I across all 97 transversals and across all 2^{13} subhypergraphs yields Parts (1)–(3). □

Over all 21 ordinary two-valued states $s \in \mathcal{S}(H_{21,13})$, the sum $s(u_1) + s(u_2) + s(u_3)$ on $U_{\text{core}} = \{1, 4, 7\}$ takes values $0, 1, 2$ with multiplicities $6, 12, 3$, and on $U_{\text{cop}} = \{3, 6, 9\}$ takes values $0, 1, 2$ with multiplicities $3, 12, 6$; on the 6 extendable states $\mathcal{S}_{\text{ext}}(H_{21,13})$ it is identically 1.

D. Two-Valued Admissibility Contexts in $\mathbb{R}^3$

Can a 3-uniform hypergraph admitting a faithful orthogonal representation in $\mathbb{R}^3$ and $\mathcal{C}_3(H) \neq \varnothing$ possess a

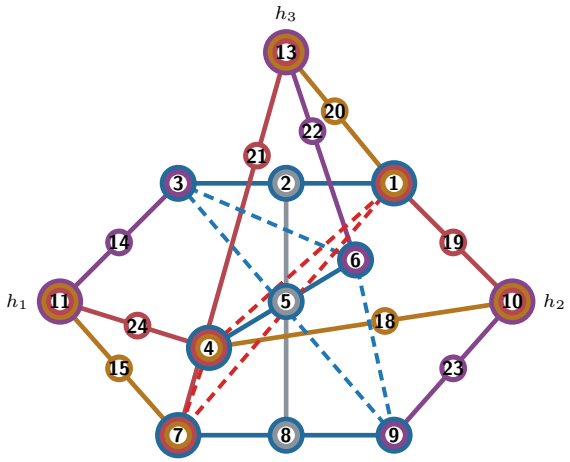


FIG. 4. The 21-ray, 13-context Yu–Oh restriction $H_{21,13}$ in $\mathbb{R}^3$ (all 13 contexts drawn as straight lines; coordinates in Table I). Highlighted are two of the 21 chromatic admissibility contexts (Theorem 7): the coplanar triple $U_{\text{cop}} = \{3, 6, 9\}$ (blue dashed triangle, $r_3 + r_6 - r_9 = \mathbf{0}$) and the 60° basis triple $U_{\text{core}} = \{1, 4, 7\}$ (red dashed triangle), both satisfying $\{c(u_1), c(u_2), c(u_3)\} = \{0, 1, 2\}$ in every 3-coloring of $H_{21,13}$.

two-valued admissibility context (Definition 1)—a triple $U = \{u_1, u_2, u_3\}$ of pairwise nonorthogonal rays whose state sum equals one across *all* admissible two-valued states $s \in \mathcal{S}(H)$, not merely on $\mathcal{S}_{\text{ext}}(H)$? By Proposition 5 and Eq. (23), any such construction must satisfy $\text{aff}\,\mathcal{S}(H) \subsetneq \mathcal{A}(H)$: $\{0, 1\}$-integrality must force state identities on $\mathcal{S}(H)$ that do not hold on the affine space $\mathcal{A}(H)$. We now establish a general roof-twin mechanism that achieves this in $\mathbb{R}^3$.

Lemma 8 (Roof-Twin and Specker-Bug Triad Lemma)**.**

1. Roof Twin Identity: *Let* $H = (V, E)$ *be a 3-uniform*

hypergraph containing two contexts $C_1 = \{a, x, y'\}$ and $C_2 = \{a, y, x'\}$ that share a roof vertex a. If both (x, y) and (x', y') are 1–0 exclusive on $\mathcal{S}(H)$,

$$s(x) + s(y) \le 1, \quad s(x') + s(y') \le 1 \quad (\forall\, s \in \mathcal{S}(H)), \tag{33}$$

then every two-valued state $s \in \mathcal{S}(H)$ satisfies the biconditional two-valued twin identities

$$s(x) = s(x'), \qquad s(y) = s(y') \qquad (\forall\, s \in \mathcal{S}(H)), \tag{34}$$

and the partially nonorthogonal triples $\{a, x, y\}$ and $\{a, x', y'\}$ satisfy

$$s(a)+s(x)+s(y) = s(a)+s(x')+s(y') = 1 \quad (\forall\, s \in \mathcal{S}(H)). \tag{35}$$

2. Universal Specker-Bug Triad Formula: *Let $u, v \in \mathbb{R}^3$ be non-collinear vectors with $\|u\| = \|v\|$ and $0 < \langle \hat{u}, \hat{v} \rangle \le \frac{1}{3}$ (after replacing v by $-v$ if needed). Set $e_1 = u + v$, $e_2 = u - v$, $e_3 = u \times v$, and*

$$\cos\phi = \sqrt{2 - \frac{\|e_1\|^2}{\|e_2\|^2}}, \qquad \sin\phi = \sqrt{\frac{\|e_1\|^2}{\|e_2\|^2} - 1}. \tag{36}$$

Then the orthogonal triad

$$\begin{aligned} \hat{f}_{1,2} &= \frac{1}{\sqrt{2}}\,\hat{e}_1 \pm \frac{1}{\sqrt{2}}\big(\cos\phi\,\hat{e}_2 + \sin\phi\,\hat{e}_3\big), \\ \hat{f}_3 &= \hat{f}_1 \times \hat{f}_2 \end{aligned} \tag{37}$$

satisfies $(f_i \times u)\cdot(f_i \times v) = 0$ ($i = 1, 2$) and generates a faithful 7-context Specker bug $B(u, v)$ in $\mathbb{R}^3$ with tips u and v. In particular, for the standard $B_{13,7}$ parametrization $r_1 = (0, 1, \alpha)$, $r_3 = (0, \alpha, -1)$, $r_7 = (1, 0, \beta)$, $r_9 = (\beta, 0, -1)$, the roof normal $r_4 = r_{10} \times r_{11}$ has squared norm

$$\|r_4\|^2 = (1 + \alpha^2)(1 + \beta^2)(1 + \alpha^2\beta^2), \tag{38}$$

which is a rational square at $(\alpha, \beta) = (\frac{9}{32}, \frac{24}{11})$ ($\|r_4\| = \frac{45305}{15488}$), so the 90° rotation around $\hat{r}_4$ maps $B(10, 11)$ to an exact integer/rational Specker bug $B(18, 24)$ in $\mathbb{Q}^3$.

Proof. (1) Let $s \in \mathcal{S}(H)$. If $s(y) = 1$, then $s(a) = 0$ (from $C_2 = \{a, y, x'\}$) and $s(x) = 0$ (from $s(x) + s(y) \le 1$), which in $C_1 = \{a, x, y'\}$ forces $s(y') = 1$; thus $s(y) \le s(y')$. Conversely, if $s(y') = 1$, then $s(a) = 0$ (from C_1) and $s(x') = 0$ (from $s(x') + s(y') \le 1$), which in C_2 forces $s(y) = 1$; thus $s(y') \le s(y)$, proving $s(y) = s(y')$. By symmetry, $s(x) = s(x')$. Substituting Eq. (34) into $s(a) + s(x) + s(y') = 1$ and $s(a) + s(y) + s(x') = 1$ yields Eq. (35).

(2) In the basis $\{e_1, e_2, e_3\}$, $u = \frac{1}{2}(e_1 + e_2)$ and $v = \frac{1}{2}(e_1 - e_2)$. By Lagrange's identity, $(f_i \times u) \cdot (f_i \times v) = \|f_i\|^2(u\cdot v) - (f_i\cdot u)(f_i\cdot v)$. For $\hat{f}_{1,2}$ in Eq. (37), $(\hat{f}_i\cdot u)(\hat{f}_i\cdot v) = \frac{1}{8}(\|e_1\|^2 - \cos^2\phi\,\|e_2\|^2) = \frac{1}{4}(\|e_1\|^2 - \|e_2\|^2) = u\cdot v$, so $r_1' = f_1 \times u \perp r_3' = f_1 \times v$ and $r_9' = f_2 \times u \perp r_7' = f_2 \times v$. Direct expansion of $r_4 = (1, -\alpha\beta, \beta) \times (\alpha\beta, -1, -\alpha) = (\beta(\alpha^2 + 1), \alpha(\beta^2 + 1), \alpha^2\beta^2 - 1)$ gives Eq. (38). □

Theorem 9 (Two-Valued Admissibility Contexts in $\mathbb{R}^3$)**.**

1. Single-Stage Roof-Twin Gadget $G_{27,16}$ *($|V| = 27$, $|E| = 16$, $N_c = 36$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 54/82$): Attaching a 7-context Specker bug $B(18, 24)$ with tips $18, 24$ to $I_{16,9}$ yields a faithful real three-dimensional hypergraph $G_{27,16}$ (realized in $\mathbb{Z}^3$ via $(\alpha, \beta) = (\frac{9}{32}, \frac{24}{11})$ in Lemma 8(2), or at $\alpha = \beta = 1$ with $r_{102} = (1, 1, \sqrt{2})$, $r_{108} = (1, 1, -\sqrt{2})$, $r_{105} = (-1, 1, 0)$). On all 82 states $s \in \mathcal{S}(G_{27,16})$, $s(10) = s(24)$ and $s(11) = s(18)$, forcing 10 partially nonorthogonal triples—including $\{4, 10, 11\}$, $\{4, 18, 24\}$, $\{3, 14, 18\}$, $\{7, 15, 18\}$, $\{1, 19, 24\}$, and $\{9, 23, 24\}$—to sum to 1 on all of $\mathcal{S}(G_{27,16})$.*

2. Two-Stage Cascade $G_{41,25}$ and $G_{41,26}$ (Genuine Two-Valued Admissibility Context): *In $G_{27,16}$, substituting $s(11) = s(18)$ into context $E_8 = \{3, 11, 14\}$ gives $s(3)+s(14)+s(18) = 1$, and hence $s(14)+s(18) \le 1$ on all of $\mathcal{S}(G_{27,16})$. Attaching the two roof contexts $E_5 = \{14, 27, 26\}$ and $E_{16} = \{18, 28, 26\}$ over $(14, 18)$ together with a third 7-context Specker bug $B(27, 28)$ between 27 and 28 yields the faithful 41-ray, 25-context hypergraph $G_{41,25}$ ($N_c = 216$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 216/396$); adding $E_3 = \{4, 5, 105\}$ (identifying $105 = 6$) yields $G_{41,26}$ ($|V| = 41$, $|E| = 26$, $N_c = 48$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 63/107$, Fig. 5) with zero extra orthogonalities at $\alpha = \beta = 1$, where*

$$\begin{aligned} r_{26} &= r_{14} \times r_{18} = (-1, 5, -3), \\ r_{27} &= r_{26} \times r_{14} = (8, -5, -11), \\ r_{28} &= r_{26} \times r_{18} = (-13, -5, -4), \end{aligned} \tag{39}$$

have $|\langle \hat{r}_{27}, \hat{r}_{28} \rangle| = \frac{35}{210} = \frac{1}{6} < \frac{1}{3}$, with central bug triad $r_{202,208} = \sqrt{35}\,(3, 0, -1) \pm \sqrt{15}\,(1, 2, 3) \pm 2\,(-1, 5, -3)$ and $r_{205} = r_{202} \times r_{208}$. By Lemma 8(1), every two-valued state $s \in \mathcal{S}(G_{41,25})$ (and $s \in \mathcal{S}(G_{41,26})$) satisfies

$$s(3) = s(26), \quad s(11) = s(18) = s(27), \quad s(14) = s(28). \tag{40}$$

Consequently, replacing $11 \mapsto 27$ and $14 \mapsto 28$ in $E_8 = \{3, 11, 14\}$ produces the pairwise-nonorthogonal two-valued admissibility context

$$U_{\text{TVS}} = \{3, 27, 28\}, \quad s(3) + s(27) + s(28) = 1 \ \ (\forall\, s \in \mathcal{S}), \tag{41}$$

with $\langle r_3, r_{27} \rangle = 6 \neq 0$, $\langle r_3, r_{28} \rangle = -1 \neq 0$, and $\langle r_{27}, r_{28} \rangle = -35 \neq 0$.

3. Two-Valued Exclusivity in $H_{21,13}$ and Promoting $H_{20,12}, H_{21,13}$ to Two-Valued Admissibility: *Among all $2^{13} = 8{,}192$ subhypergraphs of $H_{21,13}$, $|E| = 13$ ($H_{21,13}$ itself) is the sharp minimum possessing a pairwise-nonorthogonal triple that is two-valued exclusive on $\mathcal{S}$: the hub triple $U_{\text{hub}} = \{10, 11, 13\}$ uniquely satisfies $s(10) + s(11) + s(13) \le 1$ for all 21 states in $\mathcal{S}(H_{21,13})$ (and in H^+_{YO}, the four*

hubs $\{10,11,12,13\}$ *satisfy* $\sum_{i=10}^{13} s(i) \le 1$ *on all* 24 *states). Furthermore, in* $H_{20,12}$ *with* $\alpha = 3, \beta = \frac{1}{3}, c = 1$ *(*$|\langle \hat{r}_{18}, \hat{r}_{24}\rangle| = 3/\sqrt{209} < \frac{1}{3}$ *and* $|\langle \hat{r}_{20}, \hat{r}_{21}\rangle| = \frac{3}{10} < \frac{1}{3}$*), attaching Specker bugs* $B(18,24)$ *and* $B(20,21)$ *eliminates all nonextendable state restrictions to* $V(H_{20,12})$*, promoting both* $U_1^{(20)} = \{15,18,20\}$ *and* $U_2^{(20)} = \{19,21,24\}$ *to two-valued admissibility contexts; similarly, enforcing* 1–0 *exclusivity on* $(18,24)$*,* $(15,21)$*, and* $(19,20)$ *in* $H_{21,13}$ *reduces* $\mathcal{S}(H_{21,13})$ *to* $\mathcal{S}_{\text{ext}}(H_{21,13})$ *(*6 *states) and promotes all* 21 *chromatic admissibility contexts of Theorem 7 to two-valued admissibility contexts.*

Proof. Parts (1) and (2) follow directly from applying Lemma 8(1) first to the roof $\{4,11,24\},\{4,10,18\}$ over $(10,11)$ and $(18,24)$ in $G_{27,16}$, and second to the roof $\{14,27,26\},\{18,28,26\}$ over $(14,18)$ and $(27,28)$ in $G_{41,25}$ and $G_{41,26}$. Substituting $s(11) = s(27)$ and $s(14) = s(28)$ from Eq. (40) into the context equation $s(3) + s(11) + s(14) = 1$ of E_8 yields Eq. (41) for every $s \in \mathcal{S}$. Part (3) follows from exhaustive verification over all 2^{13} subhypergraphs of $H_{21,13}$ and over the state spaces of $H_{20,12}$ and $H_{21,13}$. □

Notice why Theorem 9(2) does not contradict Proposition 5: on $\mathcal{A}(G_{41,26})$ and $\mathcal{W}(G_{41,26})$, the maximally mixed assignment $p(v) = \frac{1}{3}$ satisfies $p(x) + p(y) = \frac{2}{3} \le 1$ on every Specker bug while giving $p(11) = p(18) = \frac{1}{3}$ without forcing $p(3) + p(27) + p(28) = 1$ on Born-rule states $p_\psi(v) = |\langle \psi, \hat{r}_v\rangle|^2$ (since $P_3 + P_{27} + P_{28} \neq I_3$). Thus $\operatorname{aff}\mathcal{S}(G_{41,26}) \subsetneq \mathcal{A}(G_{41,26})$.

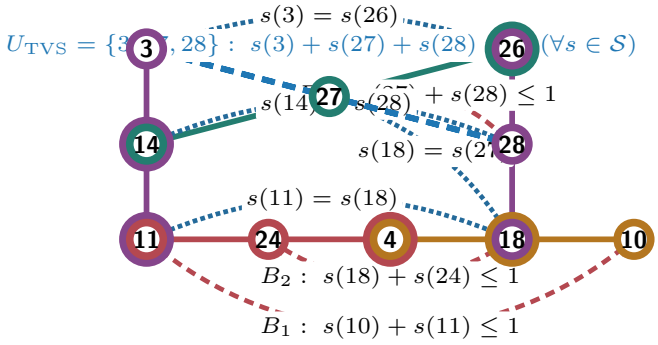


FIG. 5. Two-stage roof-twin cascade in $G_{41,25}$ and $G_{41,26}$ (Theorem 9). Stage 1 combines Specker bugs $B_1 = B(10,11)$ and $B_2 = B(18,24)$ across the roof $\{4,11,24\},\{4,10,18\}$ to force $s(11) = s(18)$ (and $s(10) = s(24)$) on all $s \in \mathcal{S}$, so context $E_8 = \{3,11,14\}$ implies $s(14) + s(18) \le 1$. Stage 2 attaches the roof $\{14,27,26\},\{18,28,26\}$ and Specker bug $B_3 = B(27,28)$, forcing $s(18) = s(27)$, $s(14) = s(28)$, and $s(3) = s(26)$ across all $s \in \mathcal{S}$. Replacing $11 \mapsto 27$ and $14 \mapsto 28$ in $E_8 = \{3,11,14\}$ yields the pairwise-nonorthogonal two-valued admissibility context $U_{\text{TVS}} = \{3,27,28\}$ (blue dashed triangle).

V. FOUR-DIMENSIONAL CHROMATIC RELATIONS IN $\mathbb{R}^4$

In dimension $d = 4$, related chromatic relations occur both in common-apex lifts and in strictly linear 4-uniform hypergraphs. The lifts $B_{14,7}^{(4)}$ and $D_{17,9}^{(4)}$ prepend a singleton apex column to every Context–Color Matrix [9, Proposition 2]; they preserve the three-dimensional universally rainbow pairs and color-multiset-equivalent pairs. The strictly linear examples ($|e \cap e'| \le 1$, girth $g \ge 3$) are $L_{18,7}^{(4)}$ ($|V| = 18$, $|E| = 7$, $N_c = 144$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 68/68$) and $L_{22,9}^{(4)}$ ($|V| = 22$, $|E| = 9$, $N_c = 216$, $|\mathcal{S}_{\text{ext}}|/|\mathcal{S}| = 92/158$) [8, 9]. The complete rainbow quadruples exhibited below are partially nonorthogonal, rather than independent admissibility contexts.

In $L_{22,9}^{(4)}$ (Fig. 6), the 9 four-element contexts are

$$\begin{aligned} C_1 &= \{1,2,4,3\}, & C_2 &= \{5,8,7,6\}, \\ C_3 &= \{4,9,10,8\}, & C_4 &= \{11,1,12,5\}, \\ C_5 &= \{11,14,13,2\}, & C_6 &= \{15,3,16,7\}, \\ C_7 &= \{15,17,18,6\}, & C_8 &= \{11,21,22,19\}, \\ C_9 &= \{13,20,15,19\}, \end{aligned} \tag{42}$$

with exact integer FOR in $\mathbb{R}^4$ given by

$$\begin{aligned} r_1 &= (1,0,0,0), & r_2 &= (0,1,0,0), \\ r_3 &= (0,0,1,0), & r_4 &= (0,0,0,1), \\ r_5 &= (0,1,-1,-1), & r_6 &= (4,-5,-11,6), \\ r_7 &= (1,-4,0,-4), & r_8 &= (4,1,1,0), \\ r_9 &= (1,-5,1,0), & r_{10} &= (2,-1,-7,0), \\ r_{11} &= (0,0,1,-1), & r_{12} &= (0,2,1,1), \\ r_{13} &= (1,0,4,4), & r_{14} &= (8,0,-1,-1), \\ r_{15} &= (4,2,0,-1), & r_{16} &= (4,-5,0,6), \\ r_{17} &= (5,-9,7,2), & r_{18} &= (2,113,77,234), \\ r_{19} &= (16,-33,-2,-2), & r_{20} &= (4,2,-21,20), \\ r_{21} &= (22,12,-11,-11), & r_{22} &= (129,44,153,153). \end{aligned} \tag{43}$$

Across all $N_c = 216$ normalized 9×4 Context–Color Matrices of $L_{22,9}^{(4)}$, the base subhypergraph $L_{18,7}^{(4)}(C_1,\ldots,C_7)$ forces the universally rainbow pair $c(11) \neq c(15)$. In $C_5 = \{11,14,13,2\}$ and $C_9 = \{13,20,15,19\}$, this propagates to 7 further nonorthogonal rainbow pairs

$$\begin{aligned} &\{11,15\},\ \{1,20\},\ \{2,20\},\ \{5,20\}, \\ &\{12,20\},\ \{14,20\},\ \{15,21\},\ \{15,22\}, \end{aligned} \tag{44}$$

the monochromatic nonorthogonal pair $11 \sim_c 20$ ($c(11) = c(20)$), and 4 partially nonorthogonal complete rainbow quadruples $(\{c(u_1), c(u_2), c(u_3), c(u_4)\} = \{0,1,2,3\}$ in all 216 normalized colorings):

$$\begin{aligned} &\{19,20,21,22\}, & &\{1,5,12,20\}, \\ &\{2,13,14,20\}, & &\{11,13,15,19\}. \end{aligned} \tag{45}$$

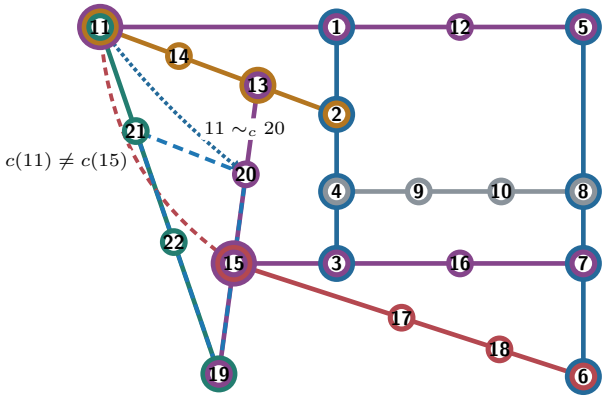


FIG. 6. The strictly linear 22-ray, 9-context hypergraph $L^{(4)}_{22,9}$ in $\mathbb{R}^4$ [Eqs. (42)–(43)], with all 9 four-element contexts drawn as straight lines. Shown are the universally rainbow pair $c(11) \neq c(15)$, the monochromatic pair $11 \sim_c 20$, and the complete rainbow quadruple $\{19, 20, 21, 22\}$ (blue dashed polygon).

VI. MASTER CENSUS AND CONCLUSION

Table II summarizes the state and coloring invariants, chromatic admissibility contexts, two-valued admissibility contexts, and related chromatic relations across all 18 faithfully coordinatized configurations studied in this work. Its paired columns omit equivalences between two already complete admissibility contexts, as specified in Definition 2.

The central phenomenon is enforced exclusivity and completeness on a single independent set, stratified into a three-tier hierarchy by the underlying state family:

1. *Chromatic admissibility contexts on $\mathcal{S}_{\text{ext}}(H)$:* Rainbow in every strong d-coloring, or equivalently normalized ($\sum_{u\in U} s(u) = 1$) on every extendable color-class state. In the Yu–Oh family, pairwise-nonorthogonal chromatic admissibility triples first appear at 12 contexts ($H_{20,12}$, 2 triples) and reach 21 triples in $H_{21,13}$ (Theorem 7), including the coplanar triple $U_{\text{cop}} = \{3, 6, 9\}$ and the 60° basis $U_{\text{core}} = \{1, 4, 7\}$.

2. *Two-valued admissibility contexts on $\mathcal{S}(H)$:* Normalized ($\sum_{u\in U} s(u) = 1$) on *every* admissible two-valued state $s \in \mathcal{S}(H)$. Even though no linear combination of context equations can force a nonorthogonal d-tuple to sum to 1 on $\mathcal{A}(H)$ (Proposition 5), $\{0, 1\}$-integrality on Specker bugs across two-context roofs enforces biconditional two-valued twins $s(x) = s(x')$ (Lemma 8), causing $\operatorname{aff}\mathcal{S}(H) \subsetneq \mathcal{A}(H)$ and yielding the pairwise-nonorthogonal two-valued admissibility context $U_{\text{TVS}} = \{3, 27, 28\}$ in $G_{41,25}$ and $G_{41,26}$ as well as 1–0-extended two-valued admissibility contexts in $H_{20,12}$ and $H_{21,13}$ (Theorem 9).

3. *Operator and affine rigidity on $\mathcal{W}(H)$ and $\mathcal{A}(H)$:* No nonorthogonal d-tuple can resolve the identity $\sum_{u\in U} P_u = I_d$ or be normalized on $\mathcal{A}(H)$. Meanwhile, balanced context certificates yield paired affine pseudocontexts ($M_{15,8}$ and H_{20}), which imply color-multiset equivalence on $\mathcal{S}_{\text{ext}}(H)$ without requiring single-set completeness.

ACKNOWLEDGMENTS

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TABLE II. Census of chromatic and two-valued admissibility contexts and related relations across 18 quantum hypergraphs in $\mathbb{R}^3$, $\mathbb{C}^3$, and $\mathbb{R}^4$. Listed are the Hilbert space $\mathbb{F}^d$, vertex/context counts, normalized coloring count N_c, extendable-to-total state ratio, universally rainbow nonorthogonal pairs, chromatic-twin pairs, chromatic admissibility contexts on $\mathcal{S}_{\text{ext}}(H)$ (with additional partially nonorthogonal complete rainbow d-tuples in parentheses), two-valued admissibility contexts on $\mathcal{S}(H)$ (with additional partially nonorthogonal two-valued complete d-tuples in parentheses), and non-twin color-multiset-equivalent pairs of sizes $k = 2, 3$.

Config.	$\mathbb{F}^d$	$(\lvert V\rvert, \lvert E\rvert)$	N_c	$\lvert\mathcal{S}_{\text{ext}}\rvert/\lvert\mathcal{S}\rvert$	Rainbow pairs	Twin pairs	Chromatic admissibility	Two-valued admissibility	Paired $k=2$	Paired $k=3$
$M_{15,8}$	$\mathbb{R}^3$	$(15, 8)$	16	24/24	0	0	0 (0)	0 (0)	0	4
H_{20}	$\mathbb{C}^3$	$(20, 12)$	12	26/26	0	0	0 (0)	0 (0)	1	0
$B_{13,7}$	$\mathbb{R}^3$	$(13, 7)$	6	12/14	1	0	0 (0)	0 (0)	0	1
$D_{16,9}$	$\mathbb{R}^3$	$(16, 9)$	7	14/21	1	0	0 (0)	0 (0)	0	0
$I_{16,9}$	$\mathbb{R}^3$	$(16, 9)$	6	12/22	10	2	0 (6)	0 (0)	2	3
$H_{17,10}$	$\mathbb{R}^3$	$(17, 10)$	4	10/18	16	2	0 (6)	0 (0)	2	3
$T_{19,11}$	$\mathbb{R}^3$	$(19, 11)$	7	14/25	8	2	0 (4)	0 (0)	1	2
$H_{20,12}$	$\mathbb{R}^3$	$(20, 12)$	4	10/25	28	4	**2** (12)	0 (0)	8	24
$H_{21,13}$	$\mathbb{R}^3$	$(21, 13)$	2	6/21	66	21	**21** (63)	0 (0)	36	0
$G_{27,16}$	$\mathbb{R}^3$	$(27, 16)$	36	54/82	18	2	0 (10)	0 (10)	4	10
$G_{41,25}$	$\mathbb{R}^3$	$(41, 25)$	216	216/396	46	6	**1** (29)	**1** (29)	10	29
$G_{41,26}$	$\mathbb{R}^3$	$(41, 26)$	48	63/107	60	6	**1** (29)	**1** (29)	10	308
$R_{21,12}$	$\mathbb{C}^3$	$(21, 12)$	32	48/54	0	0	0 (0)	0 (0)	0	12
$R_{25,15}$	$\mathbb{C}^3$	$(25, 15)$	24	48/81	0	0	0 (0)	0 (0)	0	12
$B^{(4)}_{14,7}$	$\mathbb{R}^4$	$(14, 7)$	6	13/15	1	0	0 (0)	0 (0)	0	1
$D^{(4)}_{17,9}$	$\mathbb{R}^4$	$(17, 9)$	7	15/22	1	0	0 (0)	0 (0)	0	0
$L^{(4)}_{18,7}$	$\mathbb{R}^4$	$(18, 7)$	144	68/68	1	0	0 (0)	0 (0)	0	0
$L^{(4)}_{22,9}$	$\mathbb{R}^4$	$(22, 9)$	216	92/158	8	1	0 (4)	0 (0)	0	0

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